

Практикум

E6:archive

Every question in the AA HL archive whose answer is a measurement, May 2021 to November 2025: 46 blocks, 197 marks, grouped by the eight techniques of practicum E6. No theory here and no ladder — the practicum does that. This is the topic laid out end to end, so you can see what it actually looks like on paper and how often each shape comes back.

О практикуме

Вторая половина темы

calculus.integration_applications: 46 блоков и 197 баллов, всё, где вопрос — «что этот интеграл измеряет». Разрез с E5 поимённый и идёт по вопросу: список E5 лежит в blocks карточки calculus-integration.yaml, остальное темы — сюда. Доли 0.443 и 0.557 посчитаны по баллам, а не подобраны.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-e6-areas-volumes

Собрано из ноутбука репозитория [ib-math-aa-hl-past-papers](#). Проверки в PDF не работают — они для интерактивной версии.

Archive E6: areas, volumes of revolution, accumulation

Every question in the AA HL archive whose answer is a measurement, May 2021 to November 2025: 46 blocks, 197 marks, grouped by the eight techniques of practicum E6. No theory here and no ladder — the practicum does that. This is the topic laid out end to end, so you can see what it actually looks like on paper and how often each shape comes back.

§	Technique	blocks	marks
1	Area under a curve	4	22
2	Area between two graphs	8	31
3	Volume about the x -axis	4	20
4	Volume about the y -axis	6	30
5	Washer, and volume as a condition	4	20
6	Surface of revolution	4	13
7	Displacement and distance	12	41
8	Accumulated change	4	20

How the checks work. Nothing here stores an answer to compare against. Each check measures the region, the solid, the surface or the path itself — strips, discs, frustums, tiny displacements — and compares that with what you wrote. Which is why a check can name your mistake: it knows what the displacement is, so it recognises it when you hand it in as a distance.

Three questions are asked slightly differently from the paper, and each says so where it stands: their formulas are set in the PDF in a way that does not survive text extraction, so what is asked is a step that follows from the mark scheme's own printed numbers rather than a formula I cannot read.

```
import sys
sys.path.append('.')          # from
practicum/calculus to practicum/kit.py
import sympy as sp
from kit import *

language('en')

a, b, c, h, m, n, q, r, s, w, v0 = symbols('a
b c h m n q r s w v0')

print('ready; sympy', sp.__version__)
```

1. Area under a curve

22 marks over 4 blocks, and 19 of them are Paper 1 with exact answers. The word to watch for is **total**: when the curve crosses the axis inside the limits, the area and the integral are two different numbers.

1.1 — 2021-MAY-TZ2-P1-Q10-B

5 marks · no calculator

The curve $y = 6 + 6 \cos x$ meets the x -axis nowhere on $\pi \leq x \leq 3\pi$; it sits on or above it. Show that the area of the region enclosed by the curve and the x -axis between $x = \pi$ and $x = 3\pi$ is 12π by finding it.

```
q1_1 = ...          # the area

verify_region('1.1', q1_1, 6 + 6*cos(x), 0,
pi, 3*pi)
```

1.2 — 2022-NOV-COMMON-P2-Q02-C

3 marks · calculator

The function f is defined by $f(x) = \ln(xe^x + 1) - x^4$ for $0 \leq x \leq 2$. Its graph crosses the x -axis at the origin and again at a point B .

Find the **total** area enclosed by the graph of f , the x -axis and the line $x = 2$. Give your answer to three significant figures.

```
q1_2 = ...          # the total area

verify_region('1.2', q1_2, log(x*exp(x) + 1)
- x**4, 0, 0, 2)
```

1.3 — 2023-MAY-TZ2-P1-Q12-C

7 marks · no calculator

The curve $y = \cos \sqrt{x}$ meets the x -axis where $\sqrt{x}$ is an odd multiple of $\frac{\pi}{2}$. With $x_n = \left(\frac{(2n-1)\pi}{2}\right)^2$, let R_n be the region enclosed by the curve and the x -axis for $x_n \leq x \leq x_{n+1}$.

Find the area of R_n in terms of n . (An antiderivative of $\cos \sqrt{x}$ is $2\sqrt{x} \sin \sqrt{x} + 2 \cos \sqrt{x}$.)

```
q1_3 = ...          # the area of R_n

verify_region('1.3', q1_3, cos(sqrt(x)), 0,
              ((2*n - 1)*pi/2)**2, ((2*n +
1)*pi/2)**2,
              params={n: (1, 2, 3, 4)})
```

1.4 — 2025-MAY-TZ2-P1-Q10-D

7 marks · no calculator

The function h is defined by $h(x) = \frac{4x + 1}{2x + 1}$ for $x > -\frac{1}{2}$. Let R be the region enclosed by the graph of h and the x -axis between $x = 1$ and $x = 3$.

Show that $h(x) = 2 - \frac{1}{2x + 1}$, and hence find the area of R , giving your answer in the form $p + q \ln r$.

```
q1_4 = ...          # the area of R

verify_region('1.4', q1_4, (4*x + 1)/(2*x +
1), 0, 1, 3)
```

2. Area between two graphs

31 marks over 8 blocks — the widest spread of any technique here. Seventeen marks are Paper 1, where the crossings come out exact; fourteen are Paper 2, where they come out of the GDC.

2.1 — 2021-MAY-TZ1-P2-Q10-D

5 marks · calculator

Let $f(x) = 90e^{-0.5x}$ for $x > 0$. The graph of f meets the line $y = x$ at P . The line L has gradient -1 and is tangent to the graph of f at Q ; its equation is $y = -x + 2\ln 45 + 2$.

Find the x -coordinate of the point where L meets $y = x$, and hence find the area of the region A enclosed by the graph of f , the line $y = x$ and L .

```

q2_1 = ...          # the area of A

# the region is a curvilinear triangle: the
# line  $y = x$  is on top as far as
# the crossing, the curve is on top after it,
# and  $L$  runs underneath all the
# way – so the upper boundary is a Piecewise
# and one call measures it whole
TOP = Piecewise((x, x < 5.566199), (90*exp(-
x/2), True))
verify_region('2.1', q2_1, TOP, -x +
2*log(45) + 2,
                log(45) + 1, 2*log(45),
digits=3)

```

2.2 — 2021-MAY-TZ1-P2-Q10-E

2 marks · calculator

Same f and same L . The region enclosed by the graph of f , the graph of f^{-1} and the line L is shaded.

Find its area. (Two marks, and no new integral is needed for either of them.)

```

q2_2 = ...          # the area

TOP = Piecewise((x, x < 5.566199), (90*exp(-
x/2), True))
verify_region('2.2', ... if blank(q2_2) else
q2_2/2, TOP,
                -x + 2*log(45) + 2, log(45) +
1, 2*log(45), digits=3)

```

2.3 — 2024-MAY-TZ1-P1-Q04-B

4 marks · no calculator

Let $f(x) = \cos x$ and $g(x) = \sin 2x$. Their graphs meet at $B\left(\frac{\pi}{2}, 0\right)$ and again at C , where C has x -coordinate $\frac{5\pi}{6}$.

Find the area of the region R enclosed by the two graphs between B and C .

```
q2_3 = ...          # the area of R

verify_region('2.3', q2_3, cos(x), sin(2*x),
pi/2, 5*pi/6)
```

2.4 — 2024-MAY-TZ2-P2-Q01-B

3 marks · calculator

The functions f and g are defined for $-1 \leq x \leq 0$ by $f(x) = 1 - x^2$ and $g(x) = e^{2x}$. Their graphs meet at $x = a$ and $x = b$, where $a < b$.

Find the area of the region enclosed by the two graphs.

```
q2_4 = ...          # the area

verify_region('2.4', q2_4, 1 - x**2,
exp(2*x), window=(-1, 0))
```

2.5 — 2025-MAY-TZ1-P1-Q03

4 marks · no calculator

Find the area of the region completely enclosed by the curves $y = e^x$ and $y = -e^x$ and the lines $x = -1$ and $x = 1$.

```
q2_5 = ...          # the area

verify_region('2.5', q2_5, exp(x), -exp(x),
-1, 1)
```

2.6 — 2025-MAY-TZ1-P2-Q12-E

4 marks · calculator

The function g is defined by $g(x) = \frac{1}{1 + 2x^2}$ for $0 \leq x < \frac{1}{\sqrt{2}}$, and g^{-1} exists.

The region R is completely enclosed by the curves $y = g(x)$ and $y = g^{-1}(x)$ and the two axes. Find the area of R .

```
q2_6 = ...          # the area of R

# R is symmetric in y = x: half of it is the
# region between g and that line
verify_region('2.6', ... if blank(q2_6) else
q2_6/2,
1/(1 + 2*x**2), x, 0, 0.589754,
digits=3)
```

2.7 — 2025-NOV-TZ1-P1-Q03-B

4 marks · no calculator

Find the area of the region enclosed by the line $y = -3x + 9$ and the parabola $y = -x^2 + 9$.

```
q2_7 = ...          # the area

verify_region('2.7', q2_7, -x**2 + 9, -3*x + 9)
```

2.8 — 2025-NOV-TZ3-P1-Q09-D

5 marks · no calculator

Let $f(x) = \frac{1}{2}x^2 + 5x + 13$. The line L is normal to the curve at $x = -3$, and its equation is $y = -\frac{1}{2}x + 1$.

Find the shaded area bounded by the curve, the line L and the y -axis.

```
q2_8 = ...          # the area

verify_region('2.8', q2_8, x**2/2 + 5*x + 13, -x/2 + 1, -3, 0)
```

3. Volume about the x -axis

20 marks over 4 blocks, split evenly between Paper 1 and Paper 2. On Paper 1 the square is arranged so that the integral comes out; on Paper 2 the GDC does it and the answer is three figures.

3.1 — 2021-MAY-TZ2-P2-Q11-A

6 marks · calculator

Let $f(x) = \frac{ke^{x/2}}{1 + e^x}$, where $k > 0$. The region enclosed by the graph of f , the axes and the line $x = \ln 16$ is rotated through 2π about the x -axis to model a bowl.

Show, by finding it, that the volume is $\frac{15\pi k^2}{34}$.

```
q3_1 = ...          # the volume, in terms of k

verify_solid('3.1', q3_1, k*exp(x/2)/(1 +
exp(x)), 0, log(16),
              params={k: (1, 2, 5)})
```

3.2 — 2022-NOV-COMMON-P2-Q11-C

4 marks · calculator

The function f is defined by $f(x) = e^{2x}(3x - 4)$. The region enclosed by the curve $y = f(x)$, the x -axis and the y -axis is rotated through 2π about the x -axis.

Find the volume of the solid formed, to three significant figures.

```
q3_2 = ...          # the volume

verify_solid('3.2', q3_2, exp(2*x)*(3*x - 4),
0, Rational(4, 3))
```

3.3 — 2024-MAY-TZ1-P1-Q06

6 marks · no calculator

The region R lies under $y = \sqrt{x \sin x^2}$ for $0 \leq x \leq \frac{\sqrt{\pi}}{2}$. It is rotated through 2π about the x -axis.

Show, by finding it, that the volume of the solid is $\frac{\pi(2 - \sqrt{2})}{4}$.

```
q3_3 = ...          # the volume

verify_solid('3.3', q3_3, sqrt(x*sin(x**2)),
0, sqrt(pi)/2)
```

3.4 — 2025-MAY-TZ1-P1-Q12-D

4 marks · no calculator

The region R is enclosed by $y = \cos^2 x$ and the x -axis for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. It is rotated through 2π about the x -axis.

Find the volume of the solid formed. (You may use $\int \cos^4 x \, dx = \frac{3x}{8} + \frac{\sin 2x}{4} + \frac{\sin 4x}{32}$.)

```
q3_4 = ...          # the volume

verify_solid('3.4', q3_4, cos(x)**2, -pi/2,
pi/2)
```

4. Volume about the y -axis

30 marks over 6 blocks, and 24 of them are Paper 2 — the most calculator-heavy technique of the topic after accumulation. The work is always the same first step: write x in terms of y .

4.1 — 2022-MAY-TZ1-P2-Q10-C

5 marks · calculator

Let $f(x) = \sqrt{x^2 - 1}$ for $1 \leq x \leq 2$, so that $f^{-1}(x) = \sqrt{x^2 + 1}$. The curve $y = f(x)$ is rotated 2π about the y -axis to model a water container.

(a) Show that the volume of water in the container when it is filled to height h is

$$V = \pi \left(h + \frac{h^3}{3} \right).$$

(b) Hence determine the maximum volume of the container.

```

q4_1 = ...          # (a) V in terms of h
q4_2 = ...          # (b) the maximum volume,
                    exactly

verify_solid('4.1a', q4_1, sqrt(y**2 + 1), 0,
h, var=y, axis='y',
                params={h: (1, 2, sqrt(3))})
verify_solid('4.1b', q4_2, sqrt(y**2 + 1), 0,
sqrt(3), var=y, axis='y')

```

4.2 — 2023-MAY-TZ2-P2-Q07

5 marks · calculator

Let $f(x) = \arctan(x - 2)$ for $2 \leq x \leq 2 + \sqrt{3}$.
The region bounded by the curve, the y -axis, the x -axis and the line $y = \frac{\pi}{3}$ is rotated 360° about the y -axis.

Find the volume of the solid formed, exactly.

```

q4_3 = ...          # the volume

verify_solid('4.2', q4_3, tan(y) + 2, 0,
pi/3, var=y, axis='y')

```

4.3 — 2024-MAY-TZ2-P2-Q07

5 marks · calculator

The curve $y = 4\ln(x - 2)$ for $0 \leq y \leq 4$ is rotated 360° about the y -axis.

Find the volume of the solid formed, exactly.

```
q4_4 = ...          # the volume

verify_solid('4.3', q4_4, 2 + exp(y/4), 0, 4,
var=y, axis='y')
```

4.4 — 2025-MAY-TZ2-P2-Q12-F

4 marks · calculator

The curve C has equation $4x^2 + y^2 - 24x + 4y + 20 = 0$, with a maximum point at A , where $y_A = -2 + 2\sqrt{5}$. The line $y = -4x$ touches C at B , where $y_B = -4$.

The region bounded by C , the y -axis and the lines $y = y_A$ and $y = y_B$ is rotated 360° about the y -axis. Find the volume of the solid formed, to three significant figures. (Solving C for x gives $x = 3 \pm \frac{1}{2}\sqrt{16 - 4y - y^2}$, and the region uses the left branch.)

```
q4_5 = ...          # the volume

verify_solid('4.4', q4_5, 3 - sqrt(16 - 4*y -
y**2)/2, -4, -2 + 2*sqrt(5),
            var=y, axis='y')
```

4.5 — 2025-MAY-TZ3-P1-Q07

6 marks · no calculator

The curve $x^2 + y^4 = r^4$, where $r > 0$, is drawn for $0 \leq x \leq r^2$. The region between it and the y -axis is rotated 360° about the y -axis.

Find the volume of the solid formed, in terms of r .

```
q4_6 = ...          # the volume

verify_solid('4.5', q4_6, sqrt(r**4 - y**4),
-r, r, var=y, axis='y',
            params={r: (1, 2, 3)})
```

4.6 — 2025-NOV-TZ3-P2-Q05

5 marks · calculator

The region enclosed by $y = e^{3x/4}$, the line $y = 8$ and the y -axis is rotated 360° about the y -axis.

Find the volume of the solid formed, to three significant figures.

```
q4_7 = ...          # the volume  
  
verify_solid('4.6', q4_7, 4*log(y)/3, 1, 8,  
var=y, axis='y')
```

5. Washer, and volume as a condition

20 marks over 4 blocks. Two of them subtract one solid from another; two hand you the volume and ask for a length. Same integral either way — only the last line differs.

5.1 — 2021-MAY-TZ2-P2-Q11-B

2 marks · calculator

The bowl of question 3.1 has volume $\frac{15\pi k^2}{34}$.

Find the value of k for which the bowl holds 300 cm^3 , to three significant figures.

```
q5_1 = ...      # k

verify_solid('5.1', 300, ... if blank(q5_1)
else q5_1*exp(x/2)/(1 + exp(x)),
            0, log(16), digits=3)
```

5.2 — 2022-MAY-TZ2-P2-Q06

5 marks · calculator

The curve $\frac{x^2}{36} + \frac{(y-4)^2}{16} = 1$ for $h \leq y \leq 4$ is rotated about the y -axis to form a bowl whose interior volume is 285 cm^3 .

Find the height h , to three significant figures.

```
q5_2 = ...      # h

verify_solid('5.2', 285, sqrt(36*(1 - (y -
4)**2/16))),
... if blank(q5_2) else q5_2, 4,
var=y, axis='y', digits=3)
```

5.3 — 2023-MAY-TZ1-P1-Q09

7 marks · no calculator

A sphere of radius r is formed by rotating $x = \sqrt{r^2 - y^2}$ about the y -axis. A cylindrical hole of height h is bored along the y -axis through the centre, leaving a ring.

The volume of the ring turns out not to depend on r at all. Find the value of h for which the ring has volume π .

```

q5_3 = ...          # h

verify_solid('5.3', pi, sqrt(r**2 - y**2),
            ... if blank(q5_3) else -q5_3/2,
            ... if blank(q5_3) else q5_3/2,
            inner=... if blank(q5_3) else
sqrt(r**2 - q5_3**2/4),
            var=y, axis='y', params={r: (2,
3, 5)}, digits=3)

```

5.4 — 2025-NOV-TZ1-P2-Q08

6 marks · calculator

Two functions f and g bound a region with both axes, and the region is rotated 360° about the y -axis. The mark scheme writes the first solid as $\pi \int_0^3 (y^2 + 16) dy$ and subtracts a second one from it, arriving at $V = \frac{89\pi}{2}$.

The formula for g in the question paper does not survive text extraction, and the mark scheme's transcription of it is not consistent with the diagram — so what is asked here is the part that is unambiguous. Find the volume of the first solid, the one generated by f , exactly. (Its curve

satisfies $x^2 = y^2 + 16$ and it runs from $y = 0$ to $y = 3$.)

```
q5_4 = ...          # the volume of the first
solid

verify_solid('5.4', q5_4, sqrt(y**2 + 16), 0,
3, var=y, axis='y')
```

6. Surface of revolution

All 13 marks are one Paper 3 investigation, November 2022. The formula is printed in the question; the work is making the expression under the root collapse, which it always does.

6.1 — 2022-NOV-COMMON-P3-Q02-A

2 marks · calculator

Rotating $y = mx$ for $0 \leq x \leq h$ through 360° about the x -axis gives a cone whose curved surface area is $A =$

$$2\pi \int_0^h y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

Given $m = 2$ and $h = 3$, show by finding it that $A = 18\sqrt{5}\pi$.

```
q6_1 = ...      # A
```

```
verify_surface('6.1', q6_1, 2*x, 0, 3)
```

6.2 — 2022-NOV-COMMON-P3-Q02-B-III

3 marks · calculator

Now the general cone from $y = mx$, $0 \leq x \leq h$. Its radius is $r = mh$ and its slant height is $l = h\sqrt{1 + m^2}$.

Using the same integral, find A in terms of m and h , and check for yourself that what comes out is πrl .

```
q6_2 = ...          # A in terms of m and h

verify_surface('6.2', q6_2, m*x, 0, h,
params={m: (2, 3, 1), h: (3, 1, 4)})
```

6.3 — 2022-NOV-COMMON-P3-Q02-D

4 marks · calculator

A sphere is formed by rotating the semicircle $y = \sqrt{r^2 - x^2}$, $-r \leq x \leq r$, through 360° about the x -axis.

Show by integration that its surface area is $4\pi r^2$. (The expression under the root simplifies to $\frac{r}{\sqrt{r^2 - x^2}}$, and the y in front cancels it.)

```
q6_3 = ...          # A in terms of r

verify_surface('6.3', q6_3, sqrt(r**2 -
x**2), -r, r, params={r: (2, 3, 5)})
```

6.4 — 2022-NOV-COMMON-P3-Q02-E-IV

4 marks · calculator

Let $f(x) = \sqrt{r^2 - x^2}$. The graph of $y = f(kx)$, $k > 0$, is a semi-ellipse with x -intercepts $\pm \frac{r}{k}$. Rotated 360° about the x -axis it gives an ellipsoid, whose surface area can be written as $A = 2\pi \int \sqrt{p(x)} dx$ with $p(x) = r^2 + (k^4 - k^2)x^2$.

Find A for $r = 5$ and $k = 2$, to three significant figures.

q6_4 = ... # A

```
verify_surface('6.4', q6_4, sqrt(25 -  
4*x**2), -Rational(5, 2), Rational(5, 2))
```

7. Displacement and distance

41 marks over 12 blocks — the largest technique in the topic, and nearly all of it is one distinction: $\int v$ or $\int |v|$. Nineteen marks are Paper 1, where the sign change is found exactly.

7.1 — 2021-MAY-TZ1-P2-Q04-B

2 marks · calculator

A particle moves in a straight line with velocity $v(t) = t \sin t - 3 \text{ m s}^{-1}$ for $0 \leq t \leq 10$.

Find the total distance travelled, to three significant figures.

```
q7_1 = ...          # the distance

verify_travelled('7.1', q7_1, t*sin(t) - 3,
0, 10)
```

7.2 — 2021-MAY-TZ2-P1-Q11-B

7 marks · no calculator

A particle is launched at $t = 0$ with velocity $v_0 > 0$ and moves with $v(t) = (1 + v_0)e^{-t} - 1$. It comes to rest at $t = T$, where $e^T = 1 + v_0$, and its displacement is greatest there.

Find $s_{\max}$ in terms of v_0 .

```
q7_2 = ...          # s_max

verify_position('7.2', q7_2, (1 + v0)*exp(-t)
- 1, 0, log(1 + v0),
                params={v0: (1, 3, 7)})
```

7.3 — 2021-NOV-COMMON-P1-Q10-A

7 marks · no calculator

A particle starts at the origin with velocity $v(t) = 4 + 4t - 3t^2$ m s⁻¹.

Its velocity is greatest at $t = \frac{2}{3}$. Find its displacement at that moment, exactly.

```
q7_3 = ...          # the displacement at t =
2/3

verify_position('7.3', q7_3, 4 + 4*t -
3*t**2, 0, Rational(2, 3))
```

7.4 — 2021-NOV-COMMON-P1-Q10-C

5 marks · no calculator

Same particle, $v(t) = 4 + 4t - 3t^2$. It changes direction at $t = 2$.

Find the total distance travelled in the first 3 seconds.

```
q7_4 = ...          # the distance

verify_travelled('7.4', q7_4, 4 + 4*t -
3*t**2, 0, 3)
```

7.5 — 2022-MAY-TZ1-P2-Q04-C

2 marks · calculator

A particle moves with $v(t) = e^{\sin t} + 4 \sin t$ m s⁻¹ for $0 \leq t \leq 6$. It changes direction once, at $t = 3.3469 \dots$

Find the total distance travelled, to three significant figures.

```
q7_5 = ...          # the distance

verify_travelled('7.5', q7_5, exp(sin(t)) +
4*sin(t), 0, 6)
```

7.6 — 2023-MAY-TZ2-P2-Q05-C

2 marks · calculator

A particle moves with $v(t) = 4e^{-t/3} \cos\left(\frac{t}{2} - \frac{\pi}{4}\right)$ m s⁻¹ for $0 \leq t \leq 4\pi$. Its acceleration is first zero at $t_1 = \arctan \frac{5}{12}$, and it is instantaneously at rest for the second time at $t_2 = \frac{7\pi}{2}$.

Find the distance travelled between t_1 and t_2 , to three significant figures. (The velocity changes sign once in between, at $t = \frac{3\pi}{2}$.)

```
q7_6 = ...          # the distance

verify_travelled('7.6', q7_6, 4*exp(-
t/3)*cos(t/2 - pi/4),
               atan(Rational(5, 12)),
7*pi/2)
```

7.7 — 2023-NOV-TZ1-P2-Q04-A

2 marks · calculator

A particle's displacement from a fixed point O is $s(t)$ metres, for $0 \leq t \leq 10$. It first comes to rest after q seconds.

The formula for s does not survive text extraction from the paper, so this one is asked from the mark scheme: it records $q = 5.73553\dots$

Write q to three significant figures.

```
q7_7 = ...      # q  
  
check_num('7.7', q7_7, 3, '3c45e8b47a07')
```

7.8 — 2023-NOV-TZ1-P2-Q04-B and 2023-NOV-TZ2-P2-Q04-B

6 marks · calculator, two blocks

Same particle: find the total distance travelled in the first q seconds, to three significant figures. The mark scheme records 7.68302...

This block is in the corpus twice. The November 2023 TZ1 and TZ2 papers are the same paper recorded under two codes — all twelve Paper 1 questions match line for line — so this question is counted once here and twice in the archive totals. The duplicate is worth three marks of the topic's 197.

```
q7_8 = ...          # the distance  
  
check_num('7.8', q7_8, 3, '823907ed5da3')
```

7.9 — 2024-MAY-TZ1-P2-Q04-C

2 marks · calculator

A particle passes a fixed point O at $t = 0$ and moves with $v = 2 \sin(0.5t) + 0.3t - 2$ m s⁻¹ for $0 \leq t \leq 10$.

Find its displacement relative to O when $t = 10$, to three significant figures.

```
q7_9 = ...          # the displacement  
  
verify_position('7.9', q7_9, 2*sin(0.5*t) +  
0.3*t - 2, 0, 10)
```

7.10 — 2025-MAY-TZ2-P2-Q05-B

2 marks · calculator

A particle P moves with $v(t) = e^{-\sin t} \cos 2t$ m s⁻¹ for $0 \leq t \leq 5$.

Find the total distance travelled by P , to three significant figures.

```
q7_10 = ...      # the distance

verify_travelled('7.10', q7_10, exp(-
sin(t))*cos(2*t), 0, 5)
```

7.11 — 2025-NOV-TZ1-P2-Q10-E

3 marks · calculator

An aeroplane lands 100 m in front of a stationary car, which starts moving in the same direction at that instant. Their velocities are $v_{\text{air}} = 60e^{-0.1t}$ and $v_{\text{car}} = 5t$ m s⁻¹, and the car reaches the back of the aeroplane at $t = 15.0586\dots$ s.

Find the distance the car has travelled by then, to three significant figures.

```
q7_11 = ...      # the distance

verify_amount('7.11', q7_11, 5*t, 0, at=
(15.0586088810391,))
```

7.12 — 2025-NOV-TZ3-P2-Q10-C

3 marks · calculator

A particle P has displacement $s(t) = 2^{1-t/5} \sin \frac{2\pi t}{3}$ cm from O at time t seconds.

Its displacement reaches a maximum of $1.80645\dots$, then a minimum of $-1.46729\dots$, and $s(3.5) = 1.06620\dots$

Find the total distance travelled by P in the first 3.5 seconds, to three significant figures.

```
q7_12 = ...          # the distance

# distance from a displacement function: the
# velocity is its derivative,
# and the check then walks the path in small
# steps
verify_travelled('7.12', q7_12,
                 diff(2**((1 -
t/5)*sin(2*pi*t/3), t), 0, 3.5)
```

8. Accumulated change

All 20 marks are Paper 2, and all four blocks are stories: a bowl filling, rain in a gutter, two runners, a plane and a car. The integral is the easy part; saying what accumulates is the question.

8.1 — 2022-MAY-TZ1-P2-Q10-D

2 marks · calculator

The container of question 4.1 has maximum volume $2\pi\sqrt{3}\text{ m}^3$. It starts empty and water is added at a constant $0.4\text{ m}^3\text{s}^{-1}$.

Find the time it takes to fill, exactly.

```
q8_1 = ...          # the time

verify_amount('8.1', 2*sqrt(3)*pi,
Rational(2, 5), 0, q8_1)
```

8.2 — 2023-MAY-TZ1-P2-Q10-C

5 marks · calculator

A gutter 600 cm long has cross-sectional area about 293.9 cm^2 , so it holds about

$176\,300\text{ cm}^3$. During a storm rain enters it at $R'(t) = 50 \cos \frac{2\pi t}{5} + 3000\text{ cm}^3$ per second.

Find the volume of rain that enters during a 60-second period, and so decide whether the gutter overflows.

```
q8_2 = ...          # the volume of rain in 60
seconds

verify_amount('8.2', q8_2, 50*cos(2*pi*t/5) +
3000, 0, 60)
```

8.3 — 2025-MAY-TZ3-P2-Q10-D

6 marks · calculator

Fiona and Lucy run 200 m. Fiona crosses the line at $t_f = 25.0132\dots$ s, and by then Lucy has covered $195.772\dots$ m.

Neither velocity model survives text extraction from the paper, so this one is asked from the mark scheme's own numbers: find how far Lucy is from the finishing line when Fiona completes the race, to three significant figures.

```
q8_3 = ...          # the distance from the line  
check_num('8.3', q8_3, 3, '83dc4032b069')
```

8.4 — 2025-NOV-TZ1-P2-Q10-C

7 marks · calculator

Same runway as question 7.11: $v_{\text{air}} = 60e^{-0.1t}$, $v_{\text{car}} = 5t$, and $d(t)$ is the distance between the car and the back of the aeroplane.

Given $d(0) = 100$, find $d(t)$.

```
q8_4 = ...          # d(t)  
verify_amount('8.4', q8_4, 60*exp(-t/10) -  
5*t, 0, 20, start=100)
```

What the whole topic looks like from here

197 marks, 46 blocks, 17 sessions. 78 marks on Paper 1, 106 on Paper 2, 13 on Paper 3; 119 marks carry a calculator. That is the exact mirror of E5, where 86 of 157 marks were Paper 1 — and for the

same reason from the other side: a GDC will not find you an antiderivative, and it will always find you a number.

Twelve of the 46 blocks are the same question. Given v , find how far, or find where. The only thing that changes is whether the word is *displacement* or *distance*, and 41 marks turn on reading it right.

Four blocks hand you the answer and ask for a length. The bowl that holds 300 cm^3 , the one that holds 285, the ring of volume π , the container filled at 0.4 m^3 per second. Same integral, read backwards.

One block appears twice. November 2023 TZ1 and TZ2 are one paper under two codes, and question 4(b) is counted in both. Three marks of the 197 are a second copy — worth knowing when you count how much of the topic you have actually seen.

Three blocks could not be transcribed. Their formulas are set in

the PDF in a way that breaks under text extraction, and in two of the three the mark scheme's own transcription disagrees with the diagram. Where the formula could be rebuilt from the mark scheme's numbers it was — $4e^{-t/3} \cos\left(\frac{t}{2} - \frac{\pi}{4}\right)$ and $2^{1-t/5} \sin \frac{2\pi t}{3}$ both check out to six significant figures against every value the mark scheme prints. Where it could not, the question says so.